

The EV Rebound Effect

Why cheaper journeys can mean more journeys

A plain-language guide to electric two-wheelers, rooftop solar and travel in Maharashtra

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An electric scooter can make a journey so inexpensive that you stop thinking much about its electricity bill. You may visit friends more often, accept work farther away or make a trip that you previously postponed. That change in behaviour is the starting point of this project.

The central idea: cheaper travel can increase travel. Whether that is a good outcome depends on what the extra journeys achieve, what they replace and how the electricity is supplied.

This is an explainer with worked examples, not a survey of Maharashtra households. All numerical examples are deliberately chosen assumptions. They are not current fuel quotations, electricity tariffs or measured rebound rates.

1. The same trip can feel very different

Consider a 40 km round trip. Assume petrol costs INR 105 per litre, a petrol two-wheeler travels 50 km per litre, and an electric scooter uses 0.04 kWh of electricity from the wall per kilometre. At an assumed electricity price of INR 8 per kWh:

Journey	Energy cost per km	Energy cost for 40 km
Petrol two-wheeler	INR 2.10	INR 84.00
Electric scooter	INR 0.32	INR 12.80

The electricity cost is about 85% lower in this example. The full cost of owning and using the vehicles is not 85% lower: purchase price, financing, insurance, repairs, tyres and depreciation still matter.

A rider might think: "At this price, I can afford to go." Sometimes that opens a valuable opportunity. Sometimes it produces an extra journey with little benefit. Both responses are possible.

2. What does "rebound" actually mean?

Energy-efficiency research uses **rebound** to describe behaviour or wider economic changes that take back some of the savings expected from an efficiency improvement. A technology switch to an EV also changes fuel prices, convenience and performance, so "EV-induced travel" is often the clearer umbrella term. [R1, R2]

There are three different things to track:

Measure	What it asks
More travel	Did total kilometres increase?
Savings taken back	How much of the expected energy or emissions saving was lost?
Backfire	Did the chosen impact exceed the old baseline?

These percentages are not interchangeable. Imagine a new vehicle uses one-quarter as much energy per kilometre on a consistently defined basis. At the old distance, energy use would fall from 100 units to 25 units. If distance rises by 25%, use becomes 31.25 units.

That is **25% more travel, but only 8.3% of the expected energy saving taken back**: the extra 6.25 units are 8.3% of the originally expected 75-unit saving. Total energy use remains well below the old 100-unit baseline.

The strong claim associated with **Jevons' paradox** is that efficiency ultimately increases total resource consumption. One person taking more scooter trips does not establish that claim for a household, a state or an economy.

3. Solar changes the bill, but it does not erase every cost

Rooftop solar can lower the electricity bill associated with charging. Yet a solar unit used by the scooter could sometimes have reduced another household load, earned an export payment or preserved a useful billing credit. Giving up that alternative is an **opportunity cost**.

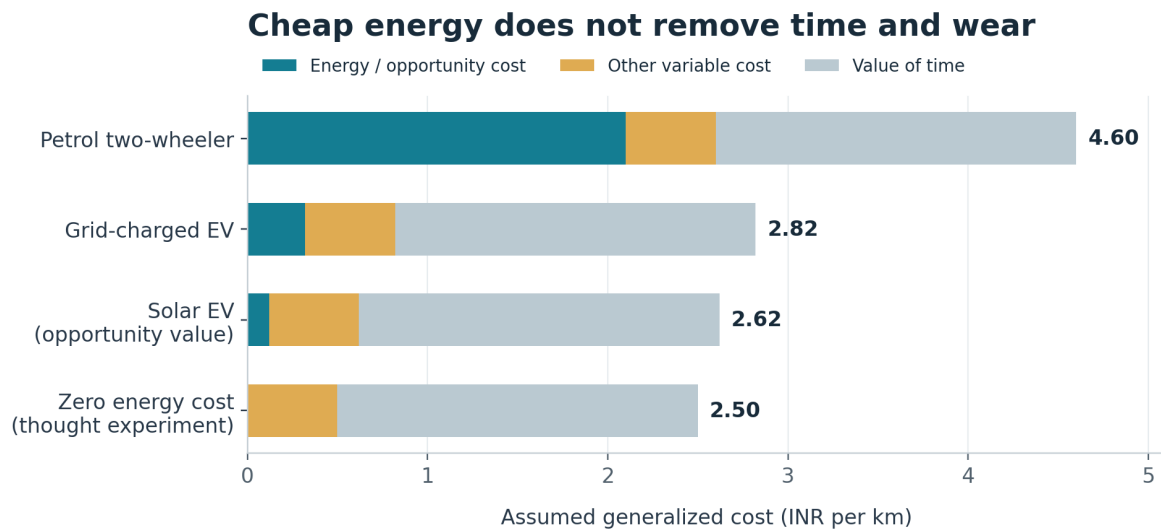
If a surplus unit would otherwise be wasted or its credit would expire unused, its opportunity cost may be very low. If it would offset an expensive grid purchase later, its value can be substantial. The answer depends on the household's actual arrangement.

Panels, inverters and vehicles also have purchase and replacement costs. Those matter when deciding whether to buy the system, even when they do not change the immediate cash cost of an additional short trip.

4. "Almost free" is still not unlimited travel

Research on zero-priced consumer products suggests that "free" can change choices beyond a small arithmetic price reduction. Whether the same response occurs when a household describes solar charging as free is a hypothesis worth testing, not an established finding about Maharashtra riders. [R3]

Time, traffic, distance, charging access and vehicle wear remain constraints. The chart adds assumed wear and time costs to the energy example. It explains why a large reduction in the electricity bill need not produce an equally large increase in travel.



Illustrative inputs only. Solar opportunity value: INR 3/kWh. Not official tariffs or measured travel costs.

How to read it: the assumed petrol energy cost falls from INR 2.10/km to INR 0.32/km with grid charging. Once the same assumed INR 2.50/km of wear and time is included, the overall decision cost falls from INR 4.60/km to INR 2.82/km, about 39%. The solar case uses an assumed opportunity value of INR 3/kWh, not an official export rate.

The time value is an analytical assumption, not money paid to anyone. Real riders will have different costs and constraints. The final bar is a thought experiment in which the energy component is zero.

5. Net metering is an accounting arrangement

With net metering, exported solar electricity can receive credits under the applicable billing rules. A credit is not a physical battery. Charging at night draws electricity supplied by the grid at that time; an annual bill offset does not prove that the charging was physically solar-powered or caused no emissions. [R4, R9]

This distinction matters when comparing daytime charging with overnight charging. A household's bill and the electricity system's emissions are related questions, but they require different calculations.

6. What the Maharashtra policy evidence supports

The following is a brief reading of specified official documents, checked on 20 September 2026. It does not calculate any household's entitlement or future bill.

Question	What the documents support
Does a larger home system always receive a larger subsidy?	The standard PM Surya Ghar household schedule shown by MSEDCL is INR 30,000/kW for the first 2 kW, plus INR 18,000 for the next kW, capped at INR 78,000. [R6]
Does a small system avoid grid-support charges?	The 2019 regulation exempts net-metering consumers with sanctioned load up to 10 kW. The March 2026 order reiterates the small-load exemption. Panel size alone is not the full test. [R4, R5]
Must I buy an EV to qualify for a bigger solar system?	The cited 2024 MSEDCL circular provides deemed feasibility approval and a load-enhancement process up to 10 kW. EV ownership is not stated as a prerequisite. [R7]

Check the applicable tariff category, sanctioned load, connection approval and billing provisions before estimating savings. Rules for large open-access projects should not be assumed to apply identically to a small residential rooftop installation.

7. Which journeys increase?

Counting the scooter's kilometres alone can be misleading.

What the EV journey replaces	What needs to be evaluated
A petrol two-wheeler or car journey	Avoided fuel use, comparable occupancy and electricity use
A bus or train journey	The service's capacity and marginal emissions, as well as fares
Walking or cycling	Additional motorised travel and changes in physical activity
Staying at home	The value of the new activity, plus the journey's costs

If a family transfers 100 km from its petrol vehicle to its EV, the EV odometer rises without increasing family vehicle-kilometres. If the family adds a new 100 km trip, total travel rises. These are different mechanisms.

New journeys can provide access to jobs, education, healthcare, customers or relationships. Leisure and social connection also have value. However, neither more kilometres nor more spending automatically proves an increase in local economic output.

8. Can emissions still fall if people travel more?

Yes, under some conditions. Suppose an EV has half the emissions per kilometre of the replaced vehicle, measured on the same basis. If distance rises by 25%, total emissions are 0.5 times 1.25, or **62.5% of the old total**. If distance doubles, that particular advantage disappears.

These are hypothetical ratios, not measurements of an Indian scooter. The result changes with the vehicle, charging electricity, manufacturing, lifetime distance and the trips displaced. A complete evaluation also counts congestion, road safety and other effects. [R9]

The better policy question is therefore: **How can cleaner, affordable transport improve people's access while keeping its wider costs low?** Useful options to evaluate include convenient daytime charging, good public transport connections and safe walking and cycling routes.

9. Turn the idea into evidence

A useful first step is a travel diary. For each journey, record distance, purpose, vehicle, occupants and what would otherwise have happened. Record charging electricity with a meter where possible, along with petrol purchases and the household's solar generation, imports and exports.

Compare several weeks before and after a change, then continue across seasons. Note changes in work, income, school schedules and the number of household vehicles. A personal diary can reveal patterns; demonstrating a causal effect requires a suitable comparison group and a stronger study design.

The accompanying academic edition sets out that research design and provides code for the illustrations. **The aim is to measure what changes, rather than assume that cheap mobility is either automatically harmful or automatically beneficial.**

Sources and project note

R1: Gillingham, Rapson & Wagner (2016), [The Rebound Effect and Energy Efficiency Policy](#).

R2: Sorrell, Dimitropoulos & Sommerville (2009), [Empirical estimates of the direct rebound effect](#).

R3: Shampanier, Mazar & Ariely (2007), [Zero as a Special Price](#).

R4: MERC (2019), [Rooftop renewable-energy regulations](#), regulations 6 and 11.

R5: MERC (2026), [Case 75 of 2025, order dated 25 March](#), paragraphs 21.12-21.14.

R6: MSEDCL, [I-SMART portal: PM Surya Ghar subsidy schedule](#).

R7: MSEDCL (2024), [Deemed approval and load enhancement up to 10 kW](#), 5 July.

R9: US EPA, [Electric Vehicle Myths](#), electricity and lifecycle discussion; not India-specific estimates.

Prepared from the author's original concept and draft with assistance from OpenAI ChatGPT for research, writing and calculations. This edition has not undergone peer review. Full references, assumptions and reproducible results accompany the academic paper.